

WaypointGen — Supplementary Material

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1 Evaluation Benchmark Details

The benchmark comprises **200 ground-truth-annotated command pairs** over 10 citrus orchards, organized as **20 command types** $\times$ **4 detail levels** (D1 global coverage, D2 multi-row patterns, D3 row segments, D4 single-tree pickup), 10 pairs per type. Command types span single-tree (pickup, inspection, replanting, via-corridor, landmark-skip), row-segment (row-then-headland, headland-then-row-half, row-central-segment, lane-then-row, row-with-inline-avoid), multi-row (every-other-tree, alternating-rows-zigzag, odd-rows-only, row-pair-corridor-transition, headland-strip-sweep), and global (quadrant-coverage, perimeter-loop, centre-then-quadrant, northern-half-then-southern-half, multi-constraint-avoidance).

2 Area-Equation Primitives

Commands compile to compositional area equations over a four-level hierarchy (L0 orchard $\rightarrow$ L1 row $\rightarrow$ L2 segment $\rightarrow$ L3 tree). Primitives: `Row(n)`, `TreePoint(x, y, r)`, `Headland(name)`, `Quadrant(dir)`, `Corridor(width_m)`, `OrchardWing`, combined with the `&` (intersection) operator and predicates (avoidance, ordering, containment, proximity). Example (D4): `Row(3).corridor(width_m=2.0) & TreePoint(tree_3_6.x, tree_3_6.y, r=1.0)`.

3 Multi-Round Inference Protocol

R1: thinking-mode inference emits the area equation + predicate set grounded in the BEV scene graph (`num_predict=2048`). R2: if structurally incomplete, re-prompt carrying R1’s reasoning trace, demanding the missing blocks. R3: deterministic geometric synthesis along the area-equation primitive as fallback. This eliminates the 24.9% missing-block failure mode of single-round inference.

4 Per-Level Results

Level	n	Struct. %	Geo. %	Both %
D1 (Global)	50	96.0	80.0	78.0
D2 (Multi-row)	50	100.0	44.0	44.0
D3 (Row-segment)	50	100.0	28.0	28.0
D4 (Single-tree)	50	100.0	6.0	6.0
Overall	200	99.0	39.5	39.0

5 3-Stage Semi-Supervised Distillation

Stage 1: 6 human-verified in-context exemplars per command type seed generation of 200 candidate trajectories. Stage 2: human-in-the-loop filtering retains the top-quality subset as weak-supervision

26 signal. Stage 3: the in-context pool is expanded with verified successes and generation is scaled
27 toward a fine-tuning corpus for a compact edge student. Round-2 generation produced 391 valid
28 area-equation samples across the 20 types on workstation- and edge-class hardware.

29 **6 Hardware**

30 Validated on three platforms: RTX 5090 (32 GB, 25.5 s/cmd), NVIDIA GB10 unified memory
31 (128 GB, 290 s/cmd), and NVIDIA Jetson Thor (122 GB unified, ~ 66 s/cmd, 99.5% structural) —
32 evidence of architecture-independent, edge-deployable inference.

33 **7 Limitations of Supplementary Scope**

34 All evaluation is open-loop (generated area equations scored against ground-truth regions); closed-
35 loop on-robot execution and a learned geometry encoder for metric fidelity are future work.